

SYSTEMS SECURITY & OPSEC SERIES

CYBERSECURITY & OPSEC THEORY & PRACTICAL FIELD GUIDE

Low-Level Network Architecture, Hardware Identifiers, Geolocation Realities, and
Threat-Model Driven VPN Analysis

PUBLISHED AND MADE BY DRAXO.DEV

SYSTEMS RESEARCH // COMPLETE CURRICULUM EDITION

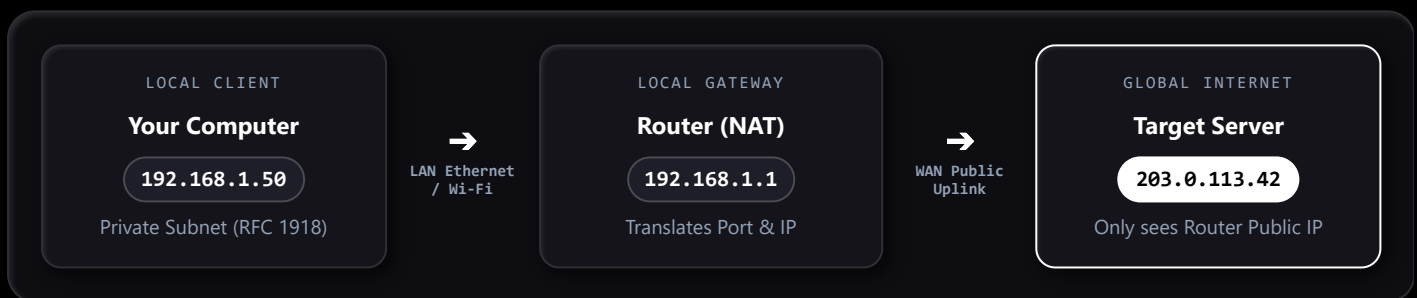
Module 01: Networking Fundamentals — IPs, Subnets, Gateways & DNS

Understanding how data moves across Layer 2 and Layer 3 of the OSI model is mandatory before talking about anonymity, privacy, or tracking. If you don't know how packets are addressed and routed, you can't properly conceal them.

■ 1. Public vs. Private IP Addresses

1.1 The Core Difference

- **Public IP:** Globally unique IP assigned to your gateway/router by your Internet Service Provider (ISP). Every device directly reachable on the public internet communicates using public IPs.
- **Private IP:** Non-routable IP addresses used strictly inside local area networks (LAN). They exist behind a router and cannot be reached directly from the wider internet without port forwarding or tunneling.



1.2 RFC 1918 Private Address Ranges

These blocks are reserved specifically for private networks:

- **Class A:** **10.0.0.0** – **10.255.255.255** (**10.0.0.0/8**) — Used in large enterprise networks, data centers, and VPN internal subnets.
- **Class B:** **172.16.0.0** – **172.31.255.255** (**172.16.0.0/12**) — Medium networks, container networks (Docker default is **172.17.0.0/16**).
- **Class C:** **192.168.0.0** – **192.168.255.255** (**192.168.0.0/16**) — Home routers and standard consumer Wi-Fi.

1.3 NAT (Network Address Translation)

Your router translates private IPs to your single public IP when you make outbound requests. It assigns an ephemeral source port on the public IP to keep track of which internal machine requested the packet.

- **Security implication:** Devices on the public internet only see your router's public IP and the assigned source port, not your internal private IP (**192.168.x.x**). However, local network adversaries can see all internal IP traffic un-NATed.

■ 2. IPv4 vs. IPv6 & The Dual-Stack Trap

2.1 Technical Comparison

FEATURE	IPV4	IPV6
Address Length	32-bit (4 octets: 192.0.2.1)	128-bit (8 hextets: 2001:0db8:85a3::8a2e:0370:7334)
Address Space	~4.29 billion (2^{32})	~ 3.4×10^{38} (2^{128})
NAT Requirement	Standard due to address exhaustion	Unnecessary; every device gets a globally unique address
Security Risk	NAT provides accidental inbound masking	Every device is directly addressable by default

2.2 The IPv6 Privacy & Tracking Nightmare

In IPv4, your whole household shares one public IP via NAT. In IPv6:

- 1. **Global Unicast Addresses (GUA):** Every single device (phone, smart TV, desktop) receives its own public, globally routable IPv6 address.
- 2. **EUI-64 MAC Leaks:** Historically, IPv6 generated the host portion of an address directly from the hardware MAC address (splitting the MAC and inserting FF:FE). This permanently tied your physical network card to your outbound IP traffic across any network.
- 3. **IPv6 Privacy Extensions (RFC 4941):** Modern OSes generate temporary, randomized IPv6 interface identifiers that rotate periodically, but misconfigured endpoints can still leak static identifiers.
- 4. **Dual-Stack VPN Bypass:** Many VPN clients only route IPv4 traffic. If your ISP supports IPv6 and your VPN does not disable or route IPv6, outbound IPv6 requests will bypass the VPN tunnel entirely and leak your real ISP-assigned IPv6 address.

■ 3. Subnets, Subnet Masks, & Default Gateways

3.1 Subnets & CIDR Notation

A subnet divides a network into smaller chunks. The **Subnet Mask** defines which bits belong to the network prefix and which belong to host addresses.

- CIDR (Classless Inter-Domain Routing) shorthand: /24 means the first 24 bits are the network ID.
- Example: 192.168.1.0/24
 - Netmask: 255.255.255.0
 - Total IP addresses: 256 (2^{32-24})
 - Usable host IPs: 254 (192.168.1.1 to 192.168.1.254)
 - Network address: 192.168.1.0
 - Broadcast address: 192.168.1.255

3.2 The Default Gateway

The gateway is the router interface connecting your local subnet to other subnets or the internet (typically 192.168.1.1 or 10.0.0.1).

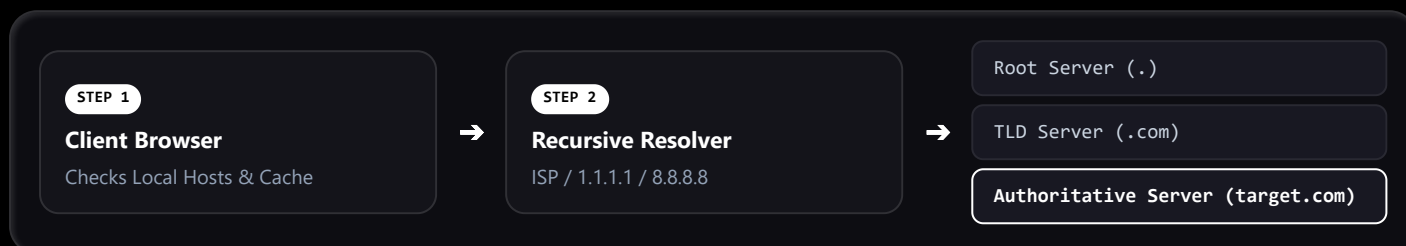
- If destination IP is in the **same subnet**: The OS resolves the target MAC address via ARP and sends frames directly across the switch/Wi-Fi.

- If destination IP is **outside the subnet**: The OS forwards the packet to the Default Gateway's MAC address, letting the gateway route it upstream.

■ 4. DNS (Domain Name System) Mechanics & Leaks

4.1 How Resolution Works

DNS converts human-readable domain names (`target.com`) into IP addresses (`93.184.216.34`).



4.2 Why Plaintext DNS Destroys Anonymity

By default, standard DNS queries run over **UDP Port 53 in plaintext**.

- Your ISP, local Wi-Fi sniffers, or upstream routers can see every domain you look up in real time, even if you connect to the site using HTTPS.
- HTTPS encrypts the HTTP payload (Layer 7), but standard DNS queries happen *before* the TLS handshake and reveal who you are contacting.

4.3 Encrypted DNS Protocols

- **DoH (DNS over HTTPS - Port 443)**: Wraps DNS queries in standard HTTPS traffic, making them indistinguishable from normal web requests to passive observers.
- **DoT (DNS over TLS - Port 853)**: Direct TLS tunnel for DNS queries. Easier for firewalls to identify and block because of the dedicated port.
- **DNSCrypt**: Encrypts traffic between the client and DNS resolver using elliptic-curve crypto.

4.4 DNS Leaks Under VPNs

A **DNS Leak** occurs when your VPN routes normal internet traffic through the encrypted VPN tunnel, but your operating system continues sending DNS queries to your ISP's DNS resolver or local router.

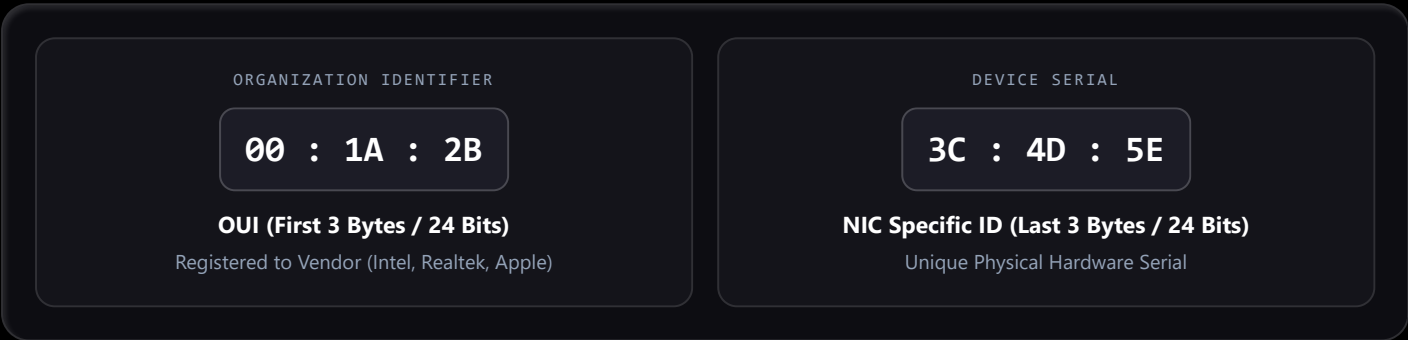
- **Cause**: Windows multi-homed name resolution (Smart Multi-Homed Name Resolution), IPv6 fallback, or DHCP-provided DNS servers overriding the virtual adapter settings.
- **Result**: An eavesdropper monitoring your ISP line sees all the sites you visit despite your active VPN.

Module 02: Hardware Identifiers — MAC Addresses & Local Footprints

Many beginners confuse MAC addresses with IP addresses and misunderstand where hardware identifiers travel. Let's break down the mechanics at Layer 2 vs Layer 3.

1. What is a MAC Address?

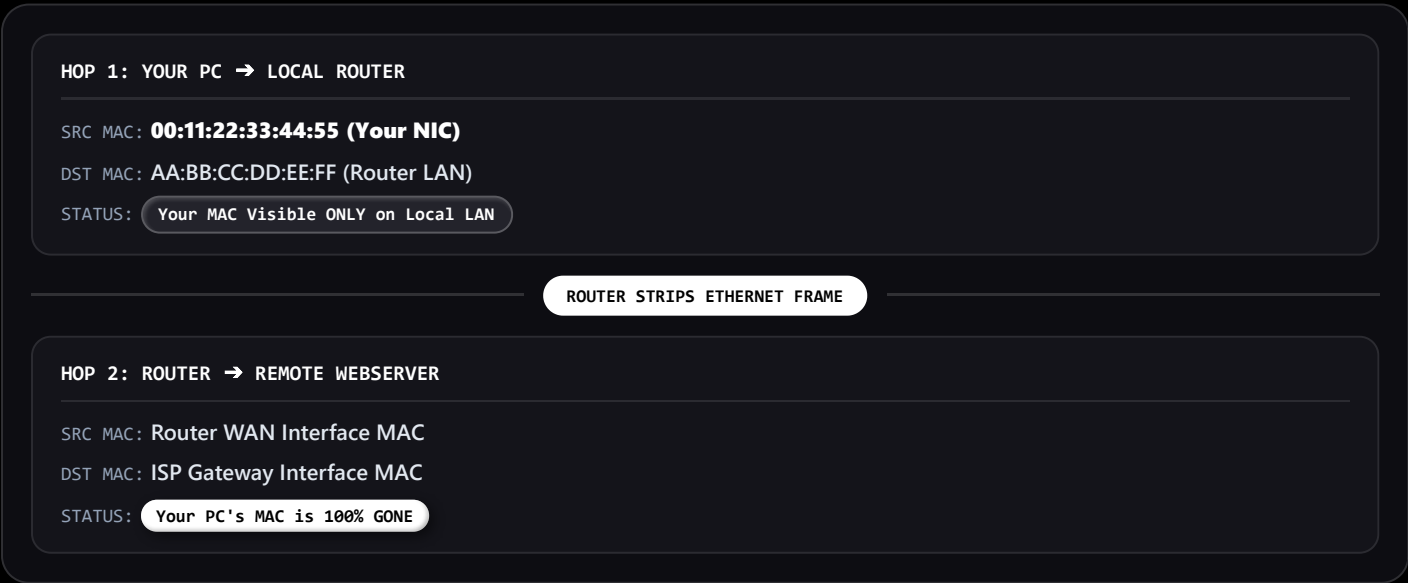
A **Media Access Control (MAC)** address is a 48-bit (6-byte) physical identifier permanently assigned to a Network Interface Card (NIC) by the manufacturer.



2. Where Does a MAC Address Travel? (The Layer 2 vs Layer 3 Rule)

2.1 The Hop-by-Hop Principle

- **MAC addresses operate exclusively at OSI Layer 2 (Data Link Layer).**
- **MAC addresses NEVER leave your local broadcast domain.**



- **Conclusion:** Remote servers, websites, and game hosts **cannot see your physical MAC address from an IP packet header.**

■ 3. How Hardware Identifiers ARE Tracked & Leaked

If MAC addresses don't leave the local network in packet headers, how do adversaries and corporations use them to track you?

3.1 Local Wi-Fi Access Points & Captive Portals

- Public Wi-Fi (airports, hotels, cafes) logs your physical MAC address the instant you associate with the Access Point (AP).
- Even if you connect to a VPN immediately after authenticating, the network operator has already tied your device's physical hardware MAC, device hostname, and connection timestamps to your physical presence.

3.2 Client-Side Software & Telemetry (Layer 7 Exfiltration)

Websites running in standard web browsers cannot directly read your MAC address due to JavaScript sandbox restrictions. However:

- **Installed Applications (Steam, Discord, Game Anti-Cheats, EDRs, Malware):** Run with native OS privileges.
- They call native APIs (e.g., Windows WinAPI `GetAdaptersAddresses()` or `GetAdaptersInfo()`) to read your real MAC address, disk serial numbers, BIOS UUID, and CPU IDs.
- They package these identifiers into encrypted HTTPS requests (Layer 7 payload) and send them to their servers, creating an unshakeable hardware profile (HWID).

3.3 ARP Sniffing on Shared Networks

On any unsegmented LAN (e.g., open Starbucks Wi-Fi or dorm networks):

- Any machine running Wireshark or `arp-scan` can discover every active host's MAC address and private IP.
- An attacker can execute **ARP Cache Poisoning (Man-in-the-Middle)** to redirect all your traffic through their machine before it reaches the router.

■ 4. MAC Spoofing & Countermeasures

4.1 MAC Randomization

Modern mobile OSes (iOS, Android) and modern Windows/Linux desktop distributions feature built-in MAC randomization for Wi-Fi scanning and per-SSID connections.

4.2 Manual Spoofing

- **Linux:**

```
sudo ip link set dev eth0 down
sudo macchanger -r eth0
sudo ip link set dev eth0 up
```

- **Windows:**

- Device Manager -> Network Adapter -> Properties -> Advanced -> `Locally Administered Address` (or `Network Address`) -> Set to a custom 12-digit hex value.
- Note: The second character must be `2`, `6`, `A`, or `E` to indicate a locally administered unicast address (e.g., `02:XX:XX:XX:XX:XX`).

Module 03: IP Tracking, Geolocation Realities & House-Level Doxxing

One of the biggest myths in internet culture and script-kiddie lore is that "pulling someone's IP gives you their exact house address." In this module, we dissect how IPs are obtained, what IP geolocation actually does, and how adversaries *actually* find exact physical locations.

■ 1. How Adversaries "Pull" Your IP Address

A remote party cannot grab your IP out of thin air; **you must establish a direct or indirect network connection with a service they control or monitor.**

1.1 Direct Peer-to-Peer (P2P) Connections

- **P2P Games & Voice Apps:** Older gaming lobbies (old Xbox 360/PS3 architecture, older GTA Online, P2P torrent swarms, legacy Skype/IRC) connect players directly to each other without dedicated relay servers.
- **Mechanism:** Running a packet capture tool like Wireshark or CommView during an active P2P session shows the direct UDP stream from your peer's IP address.
- *Modern fix:* Almost all modern platforms (Discord, modern game matchmaking) use centralized relay servers or proxy voice streams through their infrastructure, hiding user IPs behind platform server IPs.

1.2 Web Tracking Links & IP Loggers (Grabify, Canary Tokens)

- An attacker sends a link or embeds a hidden image element (``) in an HTML email or web page.
- When your client loads the asset, your browser performs a TCP handshake and HTTP GET request to the attacker's server.
- The server's access log records:
 - Your Public IP address
 - Timestamp
 - User-Agent (Browser version, OS architecture)
 - Referer header

1.3 WebRTC Leaks

- **WebRTC (Web Real-Time Communication)** allows browsers to make peer-to-peer voice/video calls.
- To punch through NAT, WebRTC queries STUN (Session Traversal Utilities for NAT) servers.
- JavaScript running on a webpage can trigger a WebRTC STUN request that exposes both your **local private IP** and your **real public IP**, even if you are using an HTTP proxy or a misconfigured VPN extension.

■ 2. The IP Geolocation Myth: Why IPs Do NOT Point to Houses

When someone enters an IP address into an online lookup tool (e.g., MaxMind GeoIP, IPinfo, DB-IP), the result is **coarse, estimated geographic routing data**, NOT a GPS coordinate of a physical building.

TARGET IP: **73.182.xx.xx**ISP / ASN: **AS7922 (Comcast)**EST. REGION: **Atlanta Metro Area**ROOFTOP ACCURACY: **0.0% (IMPOSSIBLE VIA IP)**

Country Level		99%
Region / State Level		80%
City / Metro Level		60%
Exact House / Street Address		0% (ROOFTOP IMPOSSIBLE)

2.1 How GeoIP Databases Work

- BGP and Routing Hubs:** ISPs announce IP blocks assigned to regional aggregation nodes (DSLAMs, CMTS, PoPs).
- Registry Data (RIRs):** Regional registries (ARIN, RIPE, APNIC) record the corporate mailing address of the ISP organization leasing the block, not the residential subscriber.
- Accuracy Limits:**
 - Country-level accuracy: ~95–99%
 - Region/State accuracy: ~75–90%
 - City accuracy: ~50–75%
 - Street/House accuracy: **0%** (impossible via pure IP routing tables)

■ 3. How Physical House Locations are ACTUALLY Found

If IP geolocation is inaccurate, how do skilled doxxers, threat actors, and investigators actually locate a specific home?

CRITICAL THREAT

Wi-Fi BSSID Mapping

Adversaries query nearby router MACs against war-driving databases (WiGLE/Skyhook).

Accuracy: **5 - 10 Meters**

HIGH THREAT

Breaches & OSINT

Leaked databases from food delivery, e-commerce, and billing records tied to email/phone.

Accuracy: **Exact Rooftop Address**

LEGAL / SUBPOENA

ISP DHCP Logs

ISP maintains timestamps linking IP to the customer's physical fiber/cable installation.

Accuracy: **Exact Legal Resident**

CLIENT SENSOR

Browser Geolocation

Websites trigger HTML5 GPS prompts or query onboard device positioning APIs.

Accuracy: **Sub-Meter GPS**

3.1 Wi-Fi BSSID Geolocation (War-driving Databases)

This is the most powerful remote physical location exploit:

- The Infrastructure:** Companies (Google, Apple, Skyhook) and war-driving projects (WiGLE.net) have mapped the exact physical GPS locations of billions of Wi-Fi routers worldwide.

2. **The Identifier:** Every Wi-Fi router broadcasts a unique BSSID (the MAC address of its wireless radio interface).

3. **The Exploit:**

- If an application on your machine, mobile device, or a malicious script gains access to nearby Wi-Fi network BSSIDs and their signal strengths (RSSI):
- The attacker queries the BSSID against WiGLE / Google Geolocation API.
- By trilaterating signal strengths from 2–3 nearby routers, they determine your physical rooftop location within **5 to 10 meters**.

3.2 Correlating Data Breaches & OSINT

Adversaries almost never rely solely on network packets. They chain digital footprints:

1. They obtain your username, email, or phone number from online activity.
2. They search aggregated data breach indexes (e.g., leaked databases from food delivery apps, clothing stores, credit bureaus, vehicle registrations).
3. Breaches contain full names tied to physical residential billing/shipping addresses, credit card records, and phone numbers.

3.3 ISP DHCP Logs & Legal Subpoenas

- Your ISP maintains strict **DHCP reservation logs** recording which customer account held which public IP address at any specific millisecond.
- Law enforcement with a subpoena (or corrupt ISP employees/SIM swappers with internal access) look up the IP + timestamp in the ISP's billing database to get the exact residential subscriber name, physical line installation address, and payment records.

3.4 Browser Geolocation API / Malicious Permissions

- Websites request high-precision device location via `navigator.geolocation.getCurrentPosition()`.
- If granted, the device uses onboard GPS, cell tower IDs, and Wi-Fi scanning to feed exact lat/long coordinates back to the remote web server.

Module 04: VPN Mechanics, OPSEC Realities & Provider Analysis

A Virtual Private Network (VPN) is one of the most misunderstood privacy tools in cybersecurity. Commercial marketing has convinced millions of users that clicking "Connect" makes them an untouchable ghost on the internet. In this module, we break down how VPNs actually work, where they fail, and conduct a threat-model breakdown of providers like ProtonVPN vs. Mullvad.

■ 1. How a VPN Operates Under the Hood

1.1 The Virtual Adapter & Routing Table

When a VPN client starts on your machine:

1. It creates a **virtual network interface (TUN/TAP adapter)**.
2. It modifies your operating system's **kernel routing table**, setting the default route (`0.0.0.0/0`) to point through the virtual adapter instead of your physical LAN gateway.
3. It establishes an encrypted tunnel (using protocols like **WireGuard** or **OpenVPN**) over UDP/TCP to a remote VPN server.

STANDARD CONNECTION (NO VPN)

Your App → Plaintext Headers → ISP Router (Logs Everything) → Public IP Visible → Target Website

VPN PROTECTED TUNNEL

Your App → Encrypted TUN Adapter → ISP (Sees only UDP Stream) → Decrypted at VPN Node → Target Website (Sees VPN IP)

1.2 The Trust Shift Rule

A VPN does NOT encrypt your data across the entire internet.

- The tunnel exists **only between your device and the VPN server**.
- From the VPN server to the target website, the traffic exits in whatever protocol you are using (HTTPS encrypts the payload, but the target server sees the VPN server's IP).
- **Core Rule:** A VPN does not eliminate trust; **it shifts trust from your ISP to the VPN provider.**

■ 2. Common VPN Leak Vectors & Failure Modes

Even with an active VPN, your real identity can leak through several structural flaws:

LEAK VECTOR	MECHANISM	THREAT OUTCOME
DNS Leak	OS sends DNS requests to local ISP resolver outside the VPN tunnel.	ISP still sees all domains visited.
IPv6 Leak	VPN only tunnels IPv4; outbound IPv6 requests bypass tunnel directly to ISP.	Exposes real ISP-assigned global IPv6 address.
WebRTC STUN Leak	Browser executes STUN request punching through NAT.	Remote website reads your real public and private IPs.
Kill Switch Failure	VPN connection drops momentarily; OS fails back to physical network before renegotiating.	Unencrypted packets leak real IP during reconnect.
Traffic Timing Correlation	An adversary monitoring your ISP line and the target server correlates packet burst sizes and timestamps.	Confirms you are the sender despite encryption.

3. Threat Modeling Providers: Commercial Traps vs. True OPSEC

Not all VPN providers are architected equally. When choosing an infrastructure provider, look past marketing slogans and examine the legal jurisdiction, account creation mechanics, and data retention architecture.

COMMERCIAL / PROTON

OPSEC PRIVACY SCORE

35% RATED ANONYMITY

ACCOUNT IDENTIFIER

✗ Email / Username / Password required

KYC / IDENTITY TRAIL

✗ Tied to payment records & recovery inboxes

ANONYMOUS PAYMENT

△ Limited (Requires 3rd-party KYC gateways)

LOGGING ARCHITECTURE

✗ Subject to Swiss criminal court logging orders

REAL-WORLD AUDIT

✗ Documented IP logging compliance cases

MULLVAD VPN

OPSEC PRIVACY SCORE

98% RATED ANONYMITY

ACCOUNT IDENTIFIER

✓ Random 16-Digit Token (Zero KYC)

KYC / IDENTITY TRAIL

✓ ZERO personal data collected or stored

ANONYMOUS PAYMENT

✓ Physical Cash in Mail / Monero (XMR)

LOGGING ARCHITECTURE

✓ Diskless RAM-only infrastructure

REAL-WORLD AUDIT

✓ Swedish Police raid yielded 0 logs

3.1 The Reality of Commercial VPNs & Proton

- **Account Linkage:** Providers like Proton, Nord, and Surfshark require an email address, username, and password upon registration. This immediately links your payment records, recovery accounts, and communication logs to a central identity.
- **Jurisdiction & Legal Compliance:**
 - While Switzerland (where Proton is based) has strict privacy laws, Swiss courts **can and do compel Swiss companies to log specific targets** under Swiss criminal code orders (as seen in the documented 2021 French activist case where Proton was legally forced to log IP and browser fingerprints of a target).

- No commercial provider operating a business entity is above the court orders of its sovereign jurisdiction.
- **Heavy Marketing & Affiliate Schemes:** Most "top 10 VPN" review sites are owned by the VPN companies themselves or run on massive recurring affiliate payouts, skewing technical recommendations.

3.2 Why Mullvad Sets the Benchmark for Privacy

Mullvad is widely regarded by systems engineers and security researchers as the gold standard for network proxying due to its strict zero-knowledge architecture:

1. No Account Information:

- Mullvad generates a **random 16-digit account number**.
- It never asks for an email, phone number, name, or password.

2. Anonymous Payment Methods:

- Accepts **cash mailed in a physical envelope** with only the account token written inside.
- Accepts **Monero (XMR)** for untraceable on-chain settlement.

3. RAM-Only Diskless Servers:

- Servers run on ephemeral operating system images in memory with no hard drive logging.

4. Real-World Stress Test (2023 Police Raid):

- In April 2023, the Swedish National Police visited Mullvad's Gothenburg offices with a search warrant to seize user data.
- Because of Mullvad's system architecture, no user data or logs existed to seize, and the authorities left empty-handed.

5. No Port Forwarding:

- Mullvad eliminated port forwarding to prevent users from being uniquely fingerprinted or abuse actors from compromising the network integrity.

■ 4. Advanced Tracking Beyond the VPN: Browser Fingerprinting

Using Mullvad or any VPN **only conceals your IP address**. Advanced adversaries and tracking networks do not rely on IPs alone.

4.1 What a VPN CANNOT Protect You From:

- **Browser Fingerprinting:** Websites probe your browser engine using Canvas rendering, WebGL attributes, installed fonts, audio context hardware hashing, and screen geometry. This creates a hash unique to your exact device that persists across IP changes.
- **Persistent State & Cookies:** LocalStorage, IndexedDB, Session Cookies, and Service Workers keep you authenticated and tracked regardless of what VPN node you switch to.
- **Application-Level Telemetry:** Logging into personal accounts (Google, Discord, Steam, Twitter) while connected to a VPN immediately ties that VPN exit IP to your permanent identity.

4.2 The Layered OPSEC Defense Stack

LAYER 5	Human Discipline & Zero Cross-Contamination	Never link personal accounts, phone numbers, or credit cards to burner sessions.
LAYER 4	Ephemeral Operating Systems	Tails OS / Qubes OS / Whonix inside isolated hardware hypervisors.
LAYER 3	Anti-Fingerprinting Browser Engine	Mullvad Browser / Tor Browser (Canvas & WebGL spoofing + uniform user-agents).
LAYER 2	Protocol & DNS Hardening	Encrypted DNS (DoH/DoT), WebRTC disabled, IPv6 dual-stack disabled.
LAYER 1	Network Proxy Layer	Mullvad WireGuard / Multi-hop Tor routing.

- **Golden Rule:** Anonymity is not a software you download; it is an unbroken chain of operational discipline.